

Реалізована архітектура Фінального проекту:

- **Network:** Кастомна VPC з ізольованими публічними підмережами (для веб-серверів) та приватними (для БД).
- **Compute:** Elastic Beanstalk з Load Balancer та Auto Scaling Group (масштабування за CPU > 70%).
- **Database:** Amazon RDS (MySQL), розміщена у приватних підмережах з обмеженим доступом (тільки з Security Group веб-серверів).
- **Monitoring:** Налаштовано CloudWatch Alarm для відстеження навантаження та AWS Budgets для контролю витрат.

aws

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Europe (Stockholm)

CrossFil (4516-3624-1655)

CrossFil

EC2 > Security Groups

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

Security Groups (1/4) Info

Find security groups by attribute or tag

Name

Security group ID

Security group name

VPC ID

-

[sg-0e6db319b3725db3c](#)

Database-SG

[vpc-0560996de820533a](#)

-

[sg-0d8bbd27e5689f690](#)

Web-Server-SG

[vpc-0560996de820533a](#)

-

[sg-0fa4a743d337f97fb](#)

default

[vpc-0560996de820533a](#)

-

[sg-08cac0b79407a487a](#)

default

[vpc-05e48432da8a5527](#)

sg-0d8bbd27e5689f690 - Web-Server-SG

<

Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

>

Details

Security group name

[Web-Server-SG](#)

Security group ID

[sg-0d8bbd27e5689f690](#)

Description

[Allow HTTP traffic for Elastic Beanstalk](#)

VPC ID

[vpc-0560996de820533ab](#)

Owner

451636241655

Inbound rules count

0 Permission entries

Outbound rules count

1 Permission entry

Снимок экрана

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- Target Groups
- Trust Stores
- ▼ Auto Scaling
- Auto Scaling Groups
- Settings

Security Groups (1/4) Info

 Actions ▼

Export security groups to CSV ▼

Create security group

Find security groups by attribute or tag

< 1 > 

	Name ▼	Security group ID ▼	Security group name ▼	VPC ID
☒	-	sg-0e6db319b3725db3c	Database-SG	vpc-0560996de820533a
☐	-	sg-0d8bbd27e5689f690	Web-Server-SG	vpc-0560996de820533a
☐	-	sg-0fa4a743d337f97fb	default	vpc-0560996de820533a
☐	-	sg-08cac0b79407a487a	default	vpc-05e48432da8a5527

sg-0e6db319b3725db3c - Database-SG

 ▼

< Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

>

Details

Security group name

 Database-SG

Security group ID

 sg-0e6db319b3725db3c

Description

 Allow MySQL access only fr
om Web-Server-SG

VPC ID

[vpc-0560996de820533ab](#) 

Owner

 451636241655

Inbound rules count

0 Permission entries

Outbound rules count

1 Permission entry

Aurora and RDS <

Dashboard

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Recommendations **0**

final-project-db



Modify

Actions ▼

Summary

DB identifier
final-project-db

Status
✔ Available

Role
Instance

Engine
MySQL Community

CPU
 9.00%

Class
db.t4g.micro

Current activity
 0 Connections

Region & AZ
eu-north-1b

Recommendations

< **Connectivity & security** | Monitoring | Logs & events | Configuration | Zero-ETL integrations | >

Connect using [Info](#)

☒ **Code snippets**
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ **CloudShell**
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ **Endpoints**
Use when connecting through any IDE interface.

Internet access gateway
⊖ Disabled

IAM Authentication
⊖ Disabled

Programming language
MySQL (macOS) ▼

Endpoint type
Instance endpoint ▼

Снимок экрана

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CrossFil

Elastic Beanstalk

Environments

Final-web-app-env-1

Elastic Beanstalk

Applications

Environments

Change history

▼ Application: final-web-app

Application versions

Saved configurations

▼ Environment: Final-web-app-env-1

Go to environment

Configuration

Events

Health

Logs

Monitoring

Alarms

Managed updates

Tags

Environment successfully launched.

Final-web-app-env-1

Environment overview

Health

Ok

Environment ID

e-psmxkrptmw

Domain

Final-web-app-env-1.eba-ddgmtav9.eu-north-1.elasticbeanstalk.com

Application name

final-web-app

Platform

Change version

Platform

Python 3.14 running on 64bit Amazon Linux 2023/4.9.1

Running version

-

Platform state

Supported

Events

Health

Logs

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Alarms

Managed updates

Tags

Events (20)

Filter events by text, property or value

Time

Type

Details

January 28, 2026, 11:19:19 (UTC+1)

INFO

Environment health has transitioned from Pending to Ok. Initialization completed 49 seconds ago and took 3 minutes.

CloudShell

Feedback

Console Mobile App

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Instance log streaming to CloudWatch logs

Configure the instances in your environment to stream logs to CloudWatch logs. You can set the retention to up to 10 years and configure Elastic Beanstalk to delete the logs when you terminate your environment. [Learn more](#)

Log streaming

standard CloudWatch charges apply.

☐ Enable

Environment properties

Use environment properties as either plain text environment variables or as references to secret ARN values that are stored in AWS Secrets Manager or AWS Systems Manager (SSM) Parameter Store. When using secrets, update your Elastic Beanstalk EC2 instance profile role with permissions to your secrets and parameters. [Learn more](#)

Source	Name	Value	
Plain text	PYTHONPATH	/var/app/venv/staging-LQM1lest	Remove
Plain text	DB_HOSTNAME	mysql -h final-project-db.cfem6g	Remove
Plain text	DB_PORT	3306	Remove
Plain text	DB_NAME	finaldb	Remove
Plain text	DB_USER	admin	Remove
Plain text	DB_PASSWORD	Fil24312	Remove

Add environment property

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Elastic Beanstalk

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Final-web-app-env-1

Configuration

Elastic Beanstalk

Applications

Environments

Change history

▼ Application: final-web-app

Application versions

Saved configurations

▼ Environment: Final-web-app-env-1

Go to environment

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Instance traffic and scaling

Info

Edit

Customize the capacity and scaling for your environment's instances. Select security groups to control instance traffic. Configure the software that runs on your environment's instances by setting platform-specific options.

Instances

IMDSv1

Disabled

EC2 Security Groups

sg-043362d229e5f8fb0

Capacity

Environment type

Load balanced

Min instances

1

Max instances

3

Fleet composition

On-Demand instances

On-demand base

0

On-demand above base

70

Capacity rebalancing

Disabled

Scaling cooldown

360

Processor type

x86_64

Instance types

t3.micro, t3.small

AMI ID

ami-084cac16bd83c5dab

Availability Zones

Any

Metric

CPUUtilization

Statistic

Average

Unit

Percent

Period

5

Breach duration

5

Upper threshold

70

Scale up increment

Lower threshold

Scale down increment

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Снимок экрана

Welcome to Your Elastic Beanstalk Application

Congratulations! Your Python application is now running on your own dedicated environment in the AWS Cloud.

[Learn More](#)

Benefits of AWS Elastic Beanstalk

Discover why thousands of developers rely on AWS Elastic Beanstalk to deploy and manage their applications.



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Europe (Stockholm)

CrossFil (4516-3624-1655)

CrossFil

VPC > Your VPCs > vpc-0560996de820533ab

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

vpc-0560996de820533ab / FinalProject-VPC-vpc

Actions

Details

Info

VPC ID

vpc-0560996de820533ab

DNS resolution

Enabled

Main network ACL

acl-0876f83427bb3932a

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-00d4b8a8a5b3db681

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Enabled

Main route table

rtb-0c8f5c447af780a8c

IPv6 pool

-

Owner ID

451636241655

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

Show all details

VPC

Your AWS virtual network

FinalProject-VPC-vpc

Subnets (4)

Subnets within this VPC

eu-north-1a

Route tables (4)

Route network traffic to resources

FinalProject-VPC-rtb-private1-eu-nort

CloudShell

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Encryption control mode

-

Resource map

CIDRs

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Resource map

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VPC

Your AWS virtual network

FinalProject-VPC-vpc

Subnets (4)

Subnets within this VPC

eu-north-1a

FinalProject-VPC-subnet-public1-eu-north-1a

FinalProject-VPC-subnet-private1-eu-north-1a

eu-north-1b

FinalProject-VPC-subnet-public2-eu-north-1b

FinalProject-VPC-subnet-private2-eu-north-1b

Route tables (4)

Route network traffic to resources

FinalProject-VPC-rtb-private1-eu-nort

FinalProject-VPC-rtb-public

rtb-0c8f5c447af780a8c

FinalProject-VPC-rtb-private2-eu-nort

Снимок экрана

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Global

CrossFil (4516-3624-1655)

CrossFil

Billing and Cost Management > Budgets > Budget details

Billing and Cost Management

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Getting Started

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Billing and Payments

Bills

Payments

Credits

Purchase Orders

Cost and Usage Analysis

Cost Explorer

Cost Explorer Saved Reports

Cost Anomaly Detection

Free Tier

Data Exports

Customer Carbon Footprint Tool

Cost Optimization

✔ Your budget [FinalProjectBudget](#) has been created successfully. After creating a budget, it can take up to 24 hours to populate all of your spend data.

Submit feedback

×

FinalProjectBudget [Info](#)

Delete

Edit

Budget health [Info](#)

Current vs. budgeted

0.00%

Amount spent: \$0.00 of \$1.00

Forecasted vs budgeted (MTD)

0.00%

Amount spent: - of \$1.00

Alerts [Info](#)

Thresholds

✔ OK

Actions

-

View all alerts

Details

Health status

✔ Healthy

Budget type

Cost budget [Info](#)

Budget amount

\$1.00

Period

Monthly

Start date

2026-01-01

End date

-

▶ Additional budget parameters

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CloudWatch > Alarms

CloudWatch <

Favorites and recents ▶

Ingestion

Dashboards

▼ Alarms 1 1 0

In alarm

[All alarms](#)

► AI Operations

► Application Signals (APM) [New](#)

► Infrastructure Monitoring

► Logs [New](#)

► Metrics

► Network Monitoring

► Setup

Alarms (2)

☐ Hide Auto Scaling alarms

Clear selection [Refresh](#) Create composite alarm Actions ▼ Create alarm

Search

Alarm state: Any Alarm type: Any

Actions status: Any

☐	Name	State	Last state update (UTC)	Conditions	Actions
☐	awseb-e-psmxkrptmw-stack-AWSEBCloudwatchAlarmLow-lvAcEjMCfhda	⚠ In alarm	2026-01-28 10:18:08	CPUUtilization < 20 for 1 datapoints within 5 minutes	✓ Actions enabled
☐	awseb-e-psmxkrptmw-stack-AWSEBCloudwatchAlarmHigh-ZQV1dNUHd5W	✓ OK	2026-01-28 10:17:49	CPUUtilization > 70 for 1 datapoints within 5 minutes	✓ Actions enabled

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Налаштовано автоматичний моніторинг та масштабування (Auto Scaling). Використовуються CloudWatch Alarms для відстеження завантаження процесора (CPUUtilization). При перевищенні порогу 70% система автоматично додає ресурси, а при падінні нижче 20% — оптимізує витрати, видаляючи зайві інстанси.